

Program: Diploma in Renewable Energy	
Course Code : 5299B	Course Title: Power System Lab
Semester : 5	Credits: 1.5
Course Category: Program Elective Course	
Periods per week: 3 (L: 0 T: 0 P: 3)	Periods per semester: 45

Course Objectives:

- To familiarize with various switch gears in power systems
- To apply the basic principles for operation of protective relays
- To impart the basic knowledge of measurement of earth resistance and soil resistivity
- To demonstrate the Insulation testing of LT and HT Cable

Course Prerequisites:

Topic	Course Code	Course Name	Semester
Basics theory concepts in Electrical Engineering	2031	Fundamentals of Electrical & Electronics Engineering	S2
Basic practical concepts in Electrical Engg.	2039	Fundamentals of Electrical & Electronics Engineering Lab	S2

Course Outcomes:

On completion of the course student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Select suitable switchgears for different applications.	12	Applying
CO2	Identify various protective relays and their characteristics	9	Applying
CO3	Apply suitable tests to measure the resistance of earth and soil resistivity.	9	Applying

CO4	Identify various equipments in a substation Generating Station	9	Applying
	Lab Test	6	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1				3			
CO2				3			
CO3				3			
CO4				3			

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Name of Experiment	Duration (Hours)	Cognitive Level
C01	Select suitable switchgears for different applications.		
M1.01	Identify various switchgears in the laboratory and write their specifications	3	Understanding
M1.02	Dismantle MCCB/ELCB and identify various parts	3	Applying
M1.03	Test HRC fuse by performing the load test.	3	Applying
M1.04	Test MCB by performing the load test	3	Applying
CO2	Identify various protective relays and their characteristics		
M2.01	Study different protective relays used in power system and prepare their specifications	3	Understanding

M2.02	Determine the characteristic , pick up time etc. of different types of electromagnetic relays(Eg:-Over Current and Under Voltage relays)	3	Applying
M2.03	Determine the characteristic , pick up time etc. of different types of Static/Microprocessor based relays	3	Applying
	Lab Test	3	
CO3	Apply suitable tests to measure the resistance of earth and soil resistivity.		
M3.01	Demonstrate the working of a Earth Megger	3	Applying
M3.02	Demonstrate the measurement of earth resistance and soil resistivity.	3	Applying
M3.03	Demonstrate the measurement of insulation resistance of a given LT/HT cable.	3	Applying

C04	Identify various equipments in a substation /Generating Station		
M4.01	Conduct a Visit to a near by substation / Generating station and write a report	6	Applying
M4.02	Draw a layout diagram of 11KV/400 V substation in your campus/ adjacent substation in a drawing sheet and prepare a report	3	Applying
	Lab test -2	3	

Text / Reference

T/R	Book Title/Author
T1	C.L Wadhwa ‘Electrical Power Systems’,New Age International publishers
T2	M. S. Naidu, V. Kamaraju, High Voltage Engineering, Tata McGraw-Hill Education, 2004
R1	C.L Wadhwa’high Voltage Engineering’ New Age International publishers

Online Resources

Sl.No	Website Link
1	https://electrical-engineering-portal.com/download-center/books-and-guides/relays/protection-handbook
2	https://electrical-engineering-portal.com/measuring-resistance-earth-electrode
3	https://core.ac.uk/download/pdf/234685235.pdf

Student Activity

Suggested Open-ended Experiments:

Students can do open ended experiments as a group of 3-5 preferably a simulation

work. There is no duplication in experiments in between groups. This is mainly for the purpose of continuous internal evaluation. Students should prepare a separate report on an open ended experiment of their choice.

Eg : Collect the photographs of various switchgear used in the power systems and write their specifications